Validating and Extending a Chapter-Aligned Moodle Engagement Metric Across STAT0002 and STAT0004

Evidence from online, hybrid and in-person Moodle delivery cohorts

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Abstract

Virtual learning environments such as Moodle record how students interact with course materials over time, offering a scalable basis for monitoring online behavioural engagement. A chapter-aligned, real-time engagement metric was developed and validated in prior work by Johnston et al. (2025), building on the course-wide formulation of Chong and Wong (2019). This report applies that existing metric to five module-year cohorts not used in the original validation study, from STAT0002 and STAT0004 at University College London, spanning online, hybrid and in-person delivery periods. The analysis examines association with final grades, the timing of that association, lower-tail relevance for student support, descriptive comparisons across delivery conditions, the relative contribution of Frequency, Immediacy and Diversity, alignment with different assessment components, the incremental value of temporal regularity, and latent engagement trajectories. Cumulative engagement is moderately associated with final grade across cohorts (random-effects pooled $r = 0.32$, 95% CI $[0.25, 0.38]$; mixed-effects slope ≈ 3.8 grade points per standard deviation). The association is steeper among lower-performing students, and diagnostic decomposition suggests that breadth of resource use, promptness and sustained engagement patterns carry more information than raw activity volume. Latent trajectory classes separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles. Delivery condition is perfectly confounded with academic year and cohort; comparisons across online, hybrid and in-person deliveries are therefore descriptive, and we find only limited evidence that delivery condition moderates the engagement–grade relationship.

Keywords: Moodle engagement; learning analytics; behavioural engagement; early warning; higher education; STAT0002; STAT0004.

1 Introduction

Student engagement—the time and effort students devote to learning activities—is widely linked to academic performance, satisfaction and retention. In online and blended higher education, much of the *behavioural* dimension of engagement can be observed through virtual learning environment (VLE) log data: which resources students access, how often they return, and how promptly they engage with newly released material. Moodle and similar platforms therefore provide an unobtrusive, scalable record of online study behaviour that is increasingly used in learning analytics and early student support (Henrie et al., 2015).

Institutions nonetheless face a practical tension. Threshold-based early-warning tools depend on cut-offs that are difficult to justify in advance, while predictive models often require historical outcome data and may fail to transfer across cohorts or courses. An alternative is to construct interpretable composite indicators grounded in engagement theory and computable in real time from log data alone. Johnston et al. (2025) developed such a measure by reconceptualising the course-wide engagement score of Chong and Wong (2019) into a weekly, chapter-aligned formulation based on Frequency, Immediacy and Diversity (F, I, D). Their study showed that the metric aligns with the retrospective course-wide score from early in the term, relates to final grades in structured lecture-based modules, and can help identify low-performing students through engagement quintiles—though with the modest precision typical of early-warning systems.

This report does not propose a new metric. It applies the existing chapter-aligned measure to additional cohorts from STAT0002 and STAT0004 and asks whether the association with final grades holds in a new set of module-year contexts, whether the signal emerges early enough to inform intervention, whether it is especially informative for lower-performing students, and whether richer diagnostic features—Diversity, Immediacy, temporal regularity and trajectory shape—explain more than raw engagement volume alone. Because STAT0004 spans online, hybrid and in-person deliveries across different academic years, the report also offers a descriptive comparison of how the engagement–grade relationship varies across delivery periods, while treating any such comparison as non-causal because delivery condition is confounded with cohort and year.

2 Background and metric context

Online behavioural engagement refers to observable, sustained interaction with learning activities in a digital environment. It is only a *proxy* for learning: Moodle logs capture what happens inside the VLE, not offline study, in-person attendance or the quality of students' understanding. Even so, when engagement is measured coherently and validated against academic outcomes, it can support timely monitoring and targeted support (Henrie et al., 2015).

The course-wide metric of Chong and Wong (2019) combines five indicators—Immediacy, Frequency, Diversity, Recency and Interval—into a single retrospective score aggregated across an entire course. That design is useful for end-of-term analysis but poorly suited to weekly monitoring, and it treats the course as homogeneous rather than aligned with its chapter structure. Johnston et al. (2025) addressed this by computing Frequency, Immediacy and Diversity separately for each chapter at each week, min-max scaling each indicator relative to peers at that point in the term, and summing them into a chapter-level score $\text{IDF} = F + I + D$. Chapter scores are combined across released chapters to yield a cumulative engagement measure that can be updated as the term progresses without outcome data or model training. Recency and Interval were excluded because they become unstable when measured mid-course.

Chapter alignment matters because engagement is most interpretable when measured in step with how material is released. For early-warning use, practitioners also need to understand the recall–precision trade-off: flagging the lowest-engagement students may capture many who ultimately struggle, but also many who would have succeeded without intervention (Conijn et al., 2017). The present report takes the metric definition as given and focuses on how it behaves on new UCL cohorts and what additional diagnostic insight can be extracted from its components and related temporal patterns.

3 Data

The analysis uses five module-year deliveries from two undergraduate statistics modules at University College London. STAT0002 is a first-year foundations module represented here by an online cohort (2020–21) and an in-person cohort (2022–23). STAT0004 is a second-year module and the only module in this dataset spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). Each delivery covers eleven teaching weeks with up to ten labelled chapters. Final grades are recorded on a 0–100 scale; grades below 50 indicate low performance (D/F), and below 40 indicate failure (F). Students with a recorded grade of zero, typically reflecting exam absence, were excluded.

Table 1 summarises the cohorts. The main engagement–grade sample comprises $N = 1{,}291$ students with both engagement records and valid final grades after exclusions. The latent trajectory analysis below uses $N = 1{,}290$ students with complete weekly engagement profiles suitable for class assignment; the difference of one student reflects missing weekly coverage in the trajectory model, not an error in the cohort inventory.

Table 1: Cohort inventory for the main engagement–grade sample ($N = 1{,}291$).

Module	Year	Condition	N	Mean (SD)	$\% < 50$	$\% < 40$
STAT0002	2020–21	Online	342	66.8 (13.8)	9.9	1.8
STAT0002	2022–23	In-person	268	67.4 (13.0)	10.4	2.6
STAT0004	2020–21	Online	312	62.7 (12.2)	12.5	2.9
STAT0004	2021–22	Hybrid	172	71.9 (8.8)	1.7	0.0
STAT0004	2022–23	In-person	197	70.6 (11.3)	3.6	1.0
Total			**1,291**			

Source: author's analysis of UCL Moodle log data.

Two design features constrain interpretation. First, the share of low-performing students varies widely across cohorts (from 1.7% to 12.5%), which limits how sharply any early-warning flag can perform in the highest-achieving deliveries. Second, delivery condition is **perfectly confounded** with academic year and cohort: online corresponds to 2020–21, hybrid to 2021–22, and in-person to 2022–23. Comparisons across conditions therefore describe different module-year contexts rather than randomly assigned delivery modes, and no causal claim about online, hybrid or in-person delivery can be supported.

The modules also differed in assessment structure. STAT0002 in-person records separate exam, participation and peer-marking components; STAT0004 deliveries include coursework and quiz marks, with a group component in the in-person year. This matters because Moodle engagement may align more closely with individual or exam-style assessment than with group, peer-marked or participation components. Where component marks were available, this report therefore analyses whether the engagement metric correlates more strongly with individually assessed work than with collaborative or peer-assessed components.

4 Engagement feature construction

Weekly chapter-level contribution values were available for each student. These values are stored as *per-week increments*, not as cumulative totals. For each student and week, contributions were summed across chapters to obtain weekly Frequency, Immediacy and Diversity totals. The weekly composite was then defined additively as

$$\mathrm{IDF}_{\mathrm{week}} = \mathrm{freq}_{\mathrm{week}} + \mathrm{imm}_{\mathrm{week}} + \mathrm{div}_{\mathrm{week}},$$

and cumulative measures were obtained by summing across weeks: cum_freq, cum_imm, cum_div and cum_IDF. This additive definition matches the chapter-level formulation of Johnston et al. (2025); it is not a multiplicative composite. A reconstruction check confirmed that the supplied contributions are consistent with the intended Frequency and Diversity definitions.

For extension analyses, additional timing and regularity features were derived from event logs (for example, variation in weekly activity and gaps between sessions). Latent-class mixed models were also fitted to normalised weekly cumulative IDF curves to identify distinct engagement trajectories over the term.

5 Methods

5.1 Association between engagement and final grade

Within each cohort, Spearman rank correlations were computed between cumulative engagement and final grade at each week and at the end of term. To synthesise evidence across cohorts, cohort-specific correlations were pooled using a random-effects meta-analysis, reporting the pooled correlation, confidence interval and I^2 heterogeneity statistic. Complementing rank-based evidence, a mixed-effects model was fitted with final grade as the outcome and standardised cumulative IDF as the predictor, allowing the slope to vary by cohort. This expresses the association on the grade scale in points per one standard deviation of engagement.

5.2 Timing and early-course usefulness

Weekly Spearman correlations between cumulative engagement and final grade were tracked across the term to assess how early the association emerges. For each cohort, a bootstrap procedure identified the earliest week at which the correlation was statistically indistinguishable from its end-of-term value, providing a practical sense of when the signal stabilises.

5.3 Lower-tail and early-warning analysis

Students were grouped into engagement quintiles at selected weeks, and the distribution of final grades within each quintile was examined. Quantile regression estimates the relationship between engagement and final grade at different points in the grade distribution—for example, among the lowest 10% of performers rather than only at the median—so that one can ask whether the metric matters more for students already at academic risk. Particular attention was given to the lower tail ($\tau = 0.1$) relative to the median ($\tau = 0.5$). For early-warning relevance, the bottom engagement quintile was treated as a risk flag and evaluated using the area under the ROC curve (AUC), recall and precision for identifying students scoring below 50% at the end of term.

5.4 Diagnostic decomposition of engagement

Frequency, Immediacy and Diversity are strongly correlated, so ordinary regression coefficients do not reveal which dimension carries unique information. Variance inflation factors were examined to confirm collinearity, and the LMG relative-importance decomposition was used to partition explained variance among the three indicators in a way that is stable under collinearity: it averages each predictor's contribution to R^2 over all possible orderings in which the indicators enter the model.

5.5 Assessment-component validity

Where separate assessment marks were available, engagement was correlated with each component and Steiger's test for dependent correlations was used to compare whether engagement aligns more strongly with individually assessed work (exams, quizzes) than with group- or peer-marked components.

5.6 Temporal regularity and trajectory analysis

Timing and regularity features were added to models already containing F, I and D to test whether *how* students engage over time adds information beyond *how much* and *how broadly* they engage. Separately, latent-class mixed models were applied to weekly cumulative IDF profiles to identify groups of students following distinct engagement trajectories and to compare their final grades. This treats students' weekly engagement profiles as trajectories and groups students with similar shapes of engagement over time, rather than only comparing their final cumulative engagement score.

5.7 Robustness checks

The main conclusions were examined under degree-programme adjustment, false-discovery-rate correction using the Benjamini–Hochberg procedure across weekly association tests, alternative bounded outcome specifications, and variation in the at-risk engagement threshold. These checks were kept brief and interpreted in the context of the primary findings rather than treated as a separate technical audit.

6 Results

6.1 Engagement is moderately associated with final grades across cohorts

Across all five cohorts, cumulative engagement is positively and moderately associated with final grade. The random-effects pooled rank correlation for cumulative IDF at week 11 is $r = 0.32$ (95% CI $[0.25, 0.38]$), with low-to-moderate heterogeneity across cohorts ($I^2 = 39\%$). Frequency, Immediacy and Diversity show similar pooled associations when analysed separately (Table 2). On the grade scale, the mixed-effects estimate corresponds to approximately 3.8 grade points per one standard deviation increase in engagement. Figure 1 shows that every cohort lies above zero, but the association is far from perfect prediction—engagement is a useful but imperfect proxy for academic performance.

Table 2: Pooled rank correlations and mixed-effects slopes at week 11.

Indicator	Pooled r [95% CI]	I^2	Slope (SE)	Moderator p
Frequency	0.28 [0.22, 0.34]	27%	3.36 (0.53)	0.92
Immediacy	0.30 [0.24, 0.37]	37%	3.63 (0.58)	0.88
Diversity	0.31 [0.24, 0.37]	33%	3.62 (0.53)	0.87
IDF	**0.32 [0.25, 0.38]**	**39%**	**3.79 (0.58)**	0.89

Source: author's analysis of UCL Moodle log data.

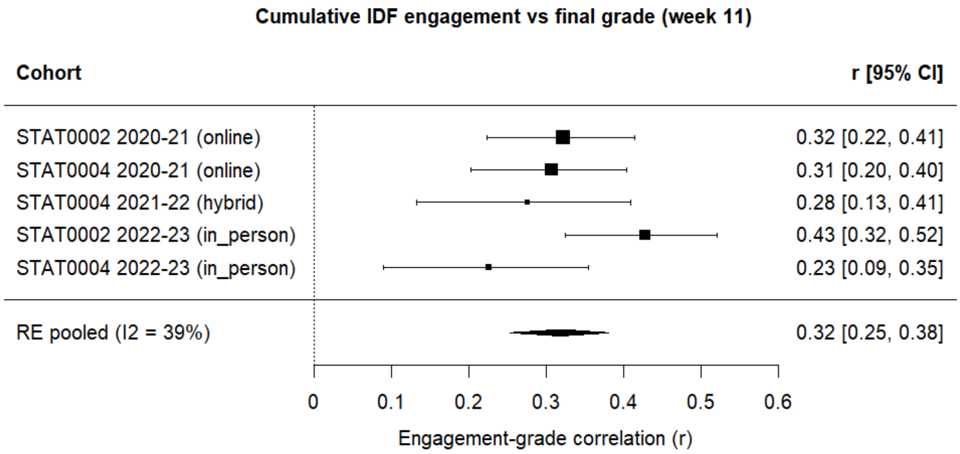


Fig. 1: Cohort-specific and pooled week-11 correlations between cumulative IDF engagement and final grade. All cohorts show positive associations; the random-effects estimate summarises the overall moderate engagement–grade relationship.

6.2 The signal appears early but not identically across cohorts

The engagement–grade association is visible from the early weeks of the term in most cohorts. Bootstrap stabilisation analysis indicates that the weekly correlation reaches its end-of-term level by week 1 or 2 in four of the five deliveries (Table 3, Figure 2). In those cohorts the signal is relatively flat across the term, so stabilisation occurs almost immediately. STAT0002 in-person is the exception: the correlation strengthens steadily and stabilises only around week 9, by which point it has become the strongest cohort-level association ($r \approx 0.41$). In practical terms, the metric already carries substantial information by the course midpoint in most deliveries, but the timing of stabilisation depends on how the signal accrues over the term.

Table 3: Stabilisation week and end-of-term IDF–grade correlation.

Cohort	Stabilisation week	End-of-term r
STAT0002 online	2	0.28
STAT0002 in-person	9	0.41
STAT0004 online	1	0.30
STAT0004 hybrid	1	0.23
STAT0004 in-person	1	0.19

Source: author's analysis of UCL Moodle log data.

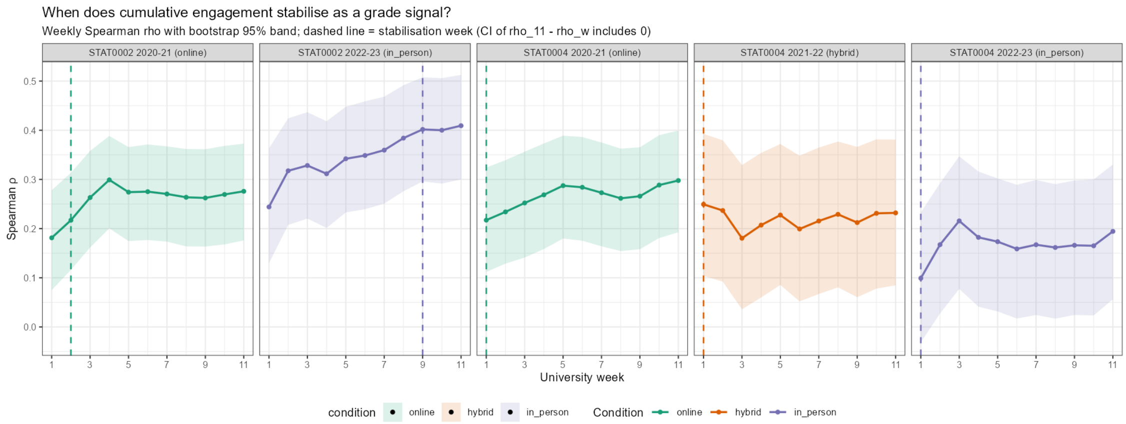


Fig. 2: Weekly Spearman correlation between cumulative engagement and final grade by cohort, with bootstrap stabilisation week marked. Most cohorts reach their end-of-term association level within the first two weeks; STAT0002 in-person shows a strengthening signal that stabilises around week 9.

6.3 Lower engagement groups tend to have lower grades

When students are ranked by cumulative engagement and grouped into quintiles, final grades tend to rise across successive quintiles, particularly in STAT0002, where quintile separation is clearest (Figure 3). At weeks 3 and 6, lowest-engagement quintile is the group whose grade distribution most often approaches or crosses the 50% low-performance threshold, while the highest-engagement quintile consistently shows the strongest outcomes. Separation between adjacent middle quintiles is weaker, which is consistent with a monotone but noisy relationship rather than sharp boundaries between moderate engagement levels.

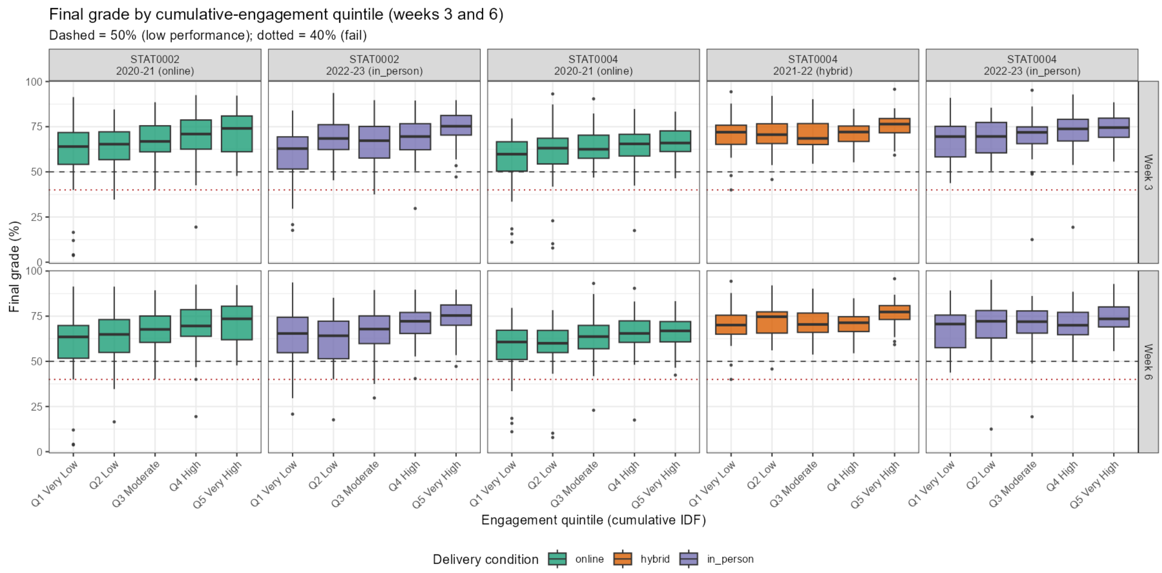


Fig. 3: Final-grade distributions by engagement quintile at weeks 3 and 6.

6.4 Engagement is especially informative in the lower grade tail

Quantile regression shows that the engagement–grade slope is steeper in the lower tail of the grade distribution than at the median in every cohort (Table 4, Figure 4). On average, the slope at $\tau = 0.1$ exceeds the median slope by about 2.3 grade points per standard deviation of engagement. This pattern suggests that variation in engagement is most consequential among students at greater academic risk, while mattering less for students who are already performing strongly—a property that supports the metric's relevance for early support, even though operational flagging still faces modest precision.

Table 4: Quantile-regression slopes (grade points per +1 SD cumulative IDF).

Cohort	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau_{0.1} - \tau_{0.5}$
STAT0002 online	5.32	3.22	2.58	2.11
STAT0002 in-person	7.88	5.04	2.96	2.84
STAT0004 online	6.06	3.15	2.17	2.91
STAT0004 hybrid	3.26	2.66	1.34	0.59
STAT0004 in-person	3.97	1.10	1.28	2.88

Source: author's analysis of UCL Moodle log data.

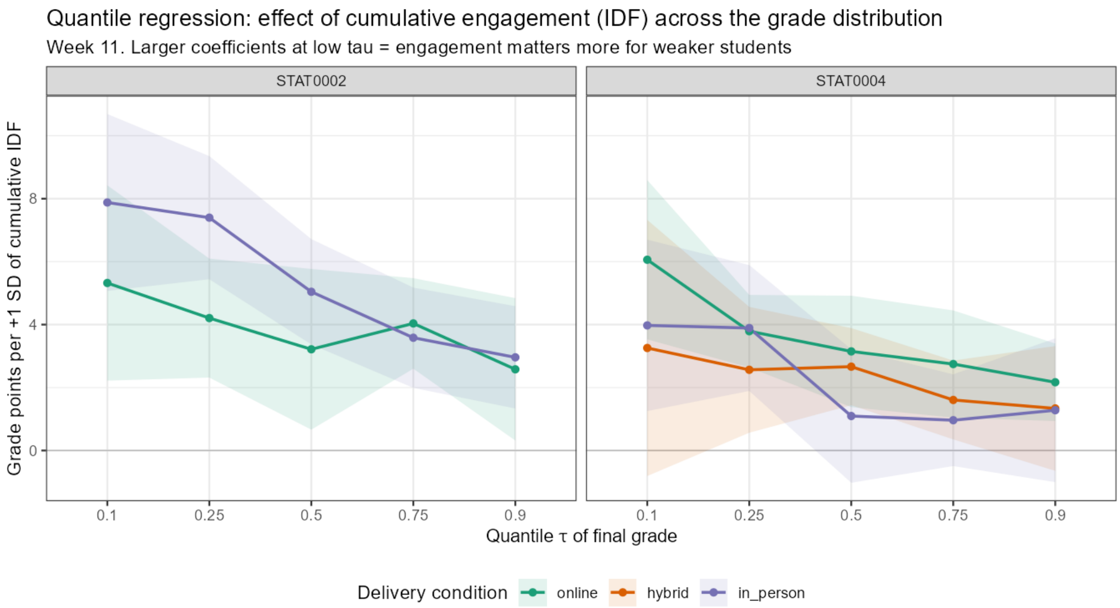


Fig. 4: Quantile-regression slopes of final grade on cumulative engagement across the grade distribution. Slopes are steeper at $\tau = 0.1$ than at the median in every cohort, indicating that engagement variation is most consequential among lower-performing students.

Treating the bottom engagement quintile as an at-risk flag at the course midpoint, STAT0002 cohorts identify roughly 38–47% of students who ultimately score below 50%, with AUC around 0.71–0.76, but precision remains modest (about 15–26%). Many flagged students still pass. Figure 5 illustrates the familiar recall–precision trade-off reported in the primary validation study.

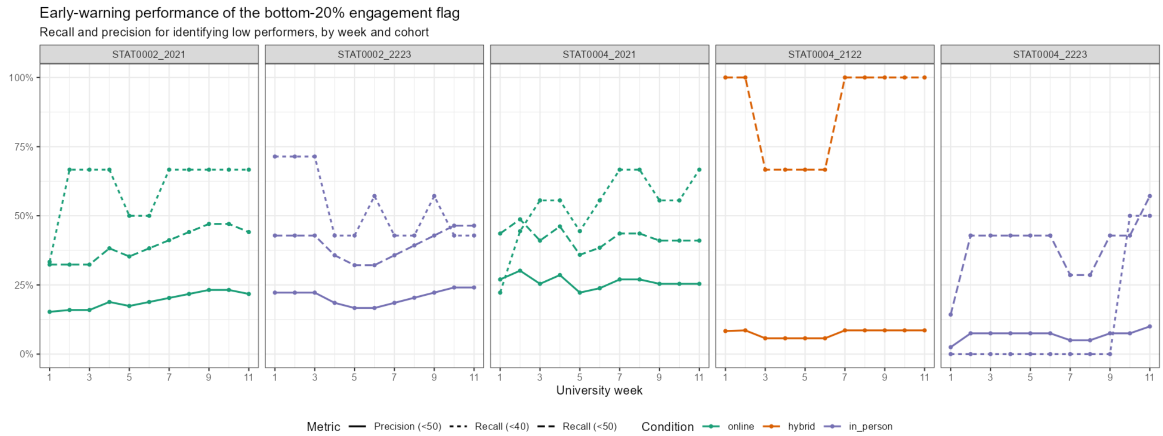


Fig. 5: Recall and precision for the bottom engagement quintile across weeks.

6.5 Delivery-condition differences are descriptive and not conclusive

Formal meta-analytic tests provide limited evidence that delivery condition explains between-cohort variation in the engagement–grade relationship (moderator $p > 0.87$ for all indicators; Table 2). Weekly correlations also appear broadly overlapping across online, hybrid and in-person deliveries (Figure 6). These patterns are consistent with a relationship that is descriptively similar across the delivery periods represented in the data. They do *not*, however, establish that delivery mode is irrelevant: because condition is confounded with academic year and cohort, the design cannot separate delivery effects from differences in students, instructors, assessment or pandemic-era context. External validation would be required before concluding that the metric needs no recalibration in a new delivery format.

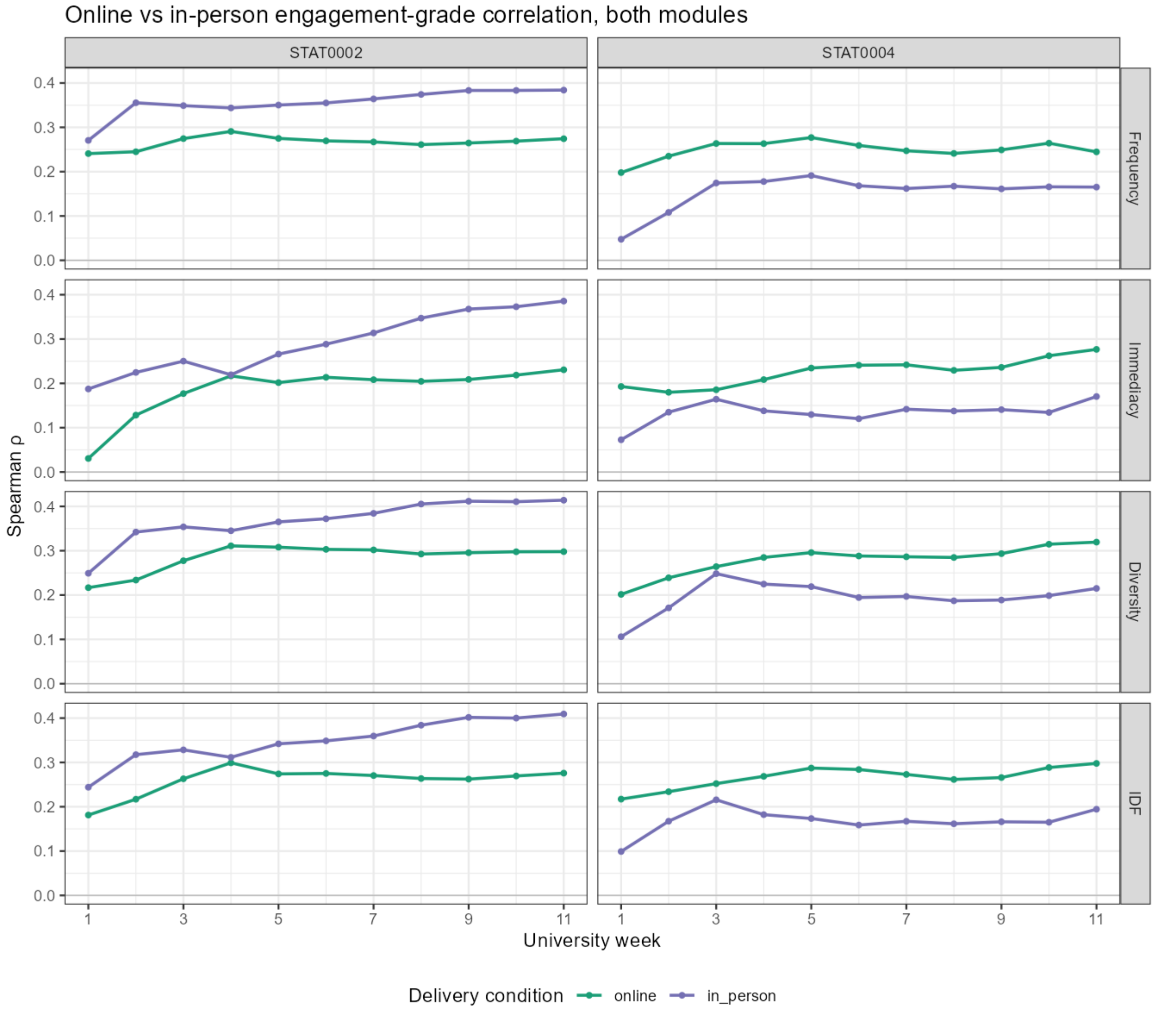


Fig. 6: Weekly engagement–grade correlation by module and delivery condition.

6.6 Breadth and promptness carry more signal than raw volume

Because Frequency, Immediacy and Diversity are highly collinear, their individual regression coefficients are difficult to interpret. The LMG decomposition provides a clearer picture (Table 5, Figure 7): Diversity typically accounts for the largest share of explained variance (about 37% on average across cohorts), Immediacy is intermediate, and Frequency the smallest (about 27%). In plain terms, students who engage with a broader range of resources and do so promptly tend to have stronger outcomes than would be expected from activity volume alone.

Table 5: LMG relative-importance shares (%) by cohort.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	28	45	27
STAT0002 in-person	28	36	36
STAT0004 online	23	32	45
STAT0004 hybrid	31	32	38
STAT0004 in-person	25	33	42

Source: author's analysis of UCL Moodle log data.

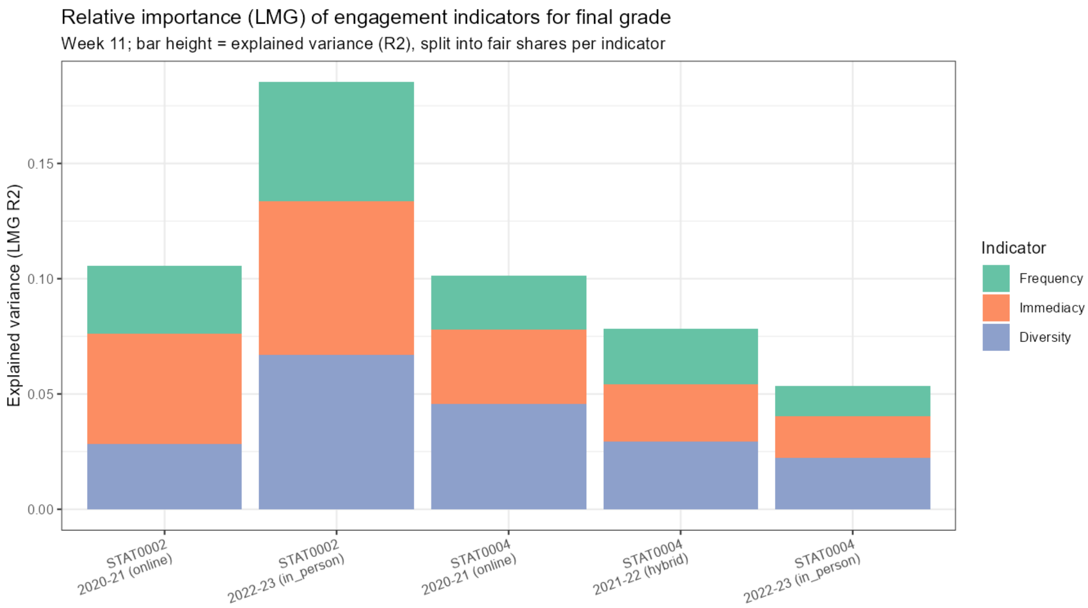


Fig. 7: LMG relative-importance shares for Frequency, Immediacy and Diversity by cohort. Diversity and Immediacy typically account for more unique explained variance than Frequency, suggesting that breadth and promptness matter more than raw activity volume.

6.7 Engagement aligns more with individual assessment than group or peer assessment

Engagement correlates more strongly with individually assessed components than with group- or peer-marked work where both are available (Figure 8). In STAT0002 in-person, the correlation with the exam grade ($r = 0.40$) is significantly larger than with peer-marking ($r = 0.20$; Steiger $z = 2.98$, $p = 0.003$). Other component contrasts point in the same direction but are not statistically significant. This supports the interpretation that Moodle engagement reflects individual study behaviour more faithfully than outcomes shaped by group dynamics or peer assessment.

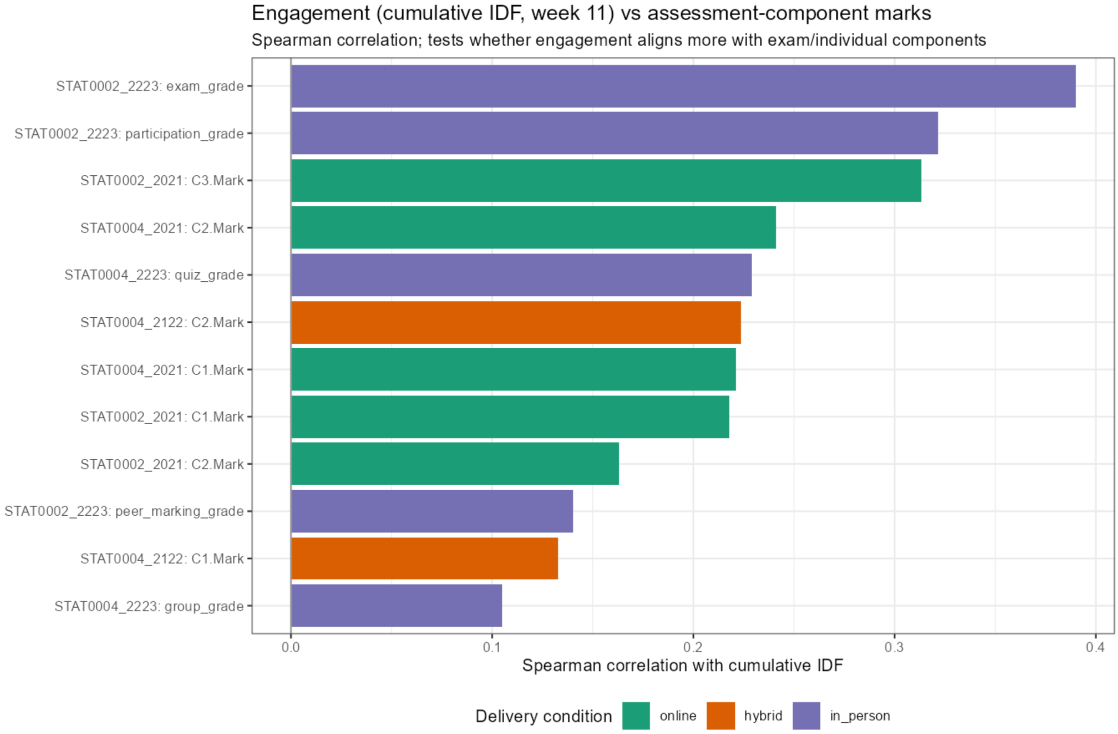


Fig. 8: Engagement–assessment component correlations by cohort.

6.8 Temporal regularity adds diagnostic value

Timing and regularity features add incremental explanatory power beyond F, I and D in the STAT0002 cohorts, where engagement showed stronger alignment with individual assessment components (Table 6, Figure 9). At week 6, the gain in R^2 is about 0.05 in both STAT0002 deliveries ($p = 0.003$ and $p = 0.006$), but not statistically significant in the STAT0004 deliveries. How regularly and promptly students work therefore appears to carry additional information in STAT0002, beyond what the core composite already captures.

Table 6: Incremental R^2 from temporal features at week 6.

Cohort	R^2 (F/I/D)	R^2 (full)	ΔR^2	p
STAT0002 online	0.091	0.143	0.052	0.003
STAT0002 in-person	0.156	0.213	0.057	0.006
STAT0004 online	0.089	0.094	0.005	0.939
STAT0004 hybrid	0.069	0.102	0.033	0.424
STAT0004 in-person	0.040	0.094	0.053	0.094

Source: author's analysis of UCL Moodle log data.

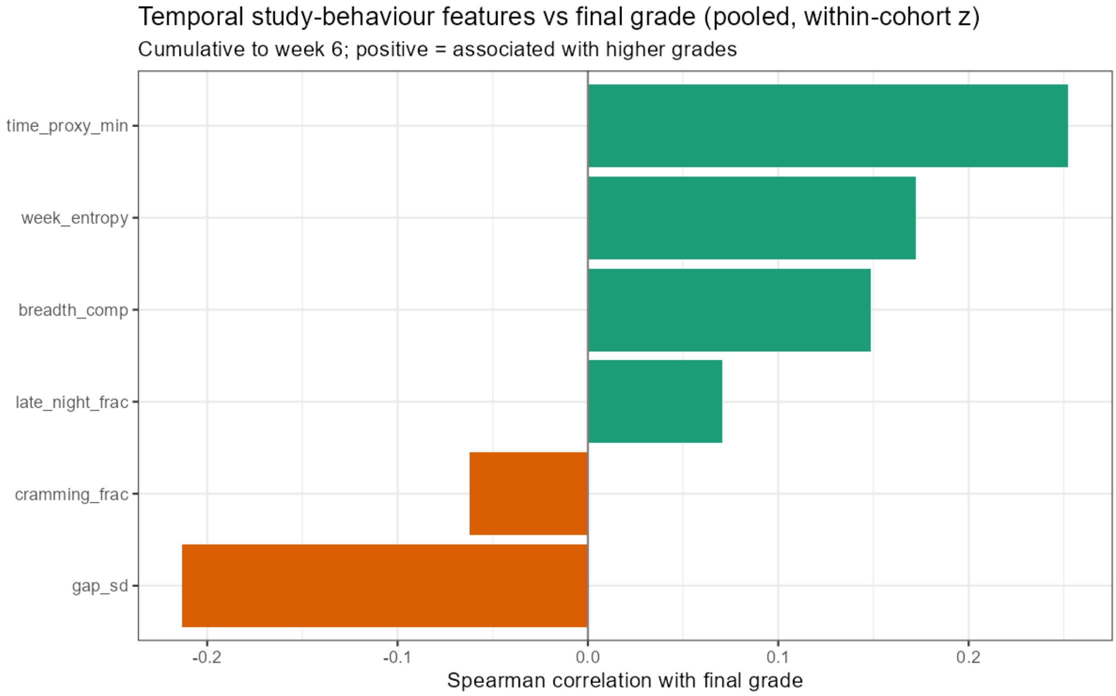


Fig. 9: Temporal and regularity features correlated with final grade.

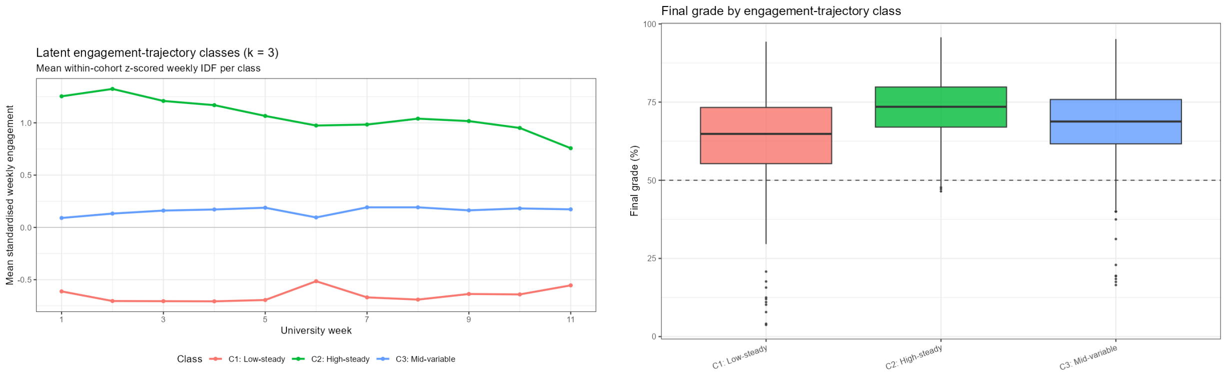
6.9 Engagement trajectories separate student outcomes

Latent-class analysis of weekly cumulative IDF profiles favours a three-class solution and identifies interpretable trajectory types: a low-steady group, a high-steady group and a larger mid-variable group (Figure 10). Final grades differ markedly across classes (Kruskal–Wallis $p = 2.6 \times 10^{-17}$). The high-steady class averages 72.9, compared with 63.2 for the low-steady class—a gap of about 9.7 grade points—and the low-steady class contains the highest share of low performers (15.7%). This is among the strongest evidence in the report that sustained engagement patterns over the term carry meaningful information about outcomes, beyond a single end-of-term score.

Table 7: Trajectory classes and grade outcomes ($N = 1{,}290$).

Class	N	Mean grade	$\% < 50$
Low-steady	439	63.2	15.7
High-steady	165	72.9	2.4
Mid-variable	686	68.3	5.5

Source: author's analysis of UCL Moodle log data.



(a) Trajectory shapes

(b) Grades by class

Fig. 10: Latent engagement trajectories (left) and final-grade distributions by trajectory class (right). Three trajectory types—low-steady, high-steady and mid-variable—separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles.

6.10 Robustness checks

The headline association is not an artefact of programme composition: adjusting for degree programme changes the engagement slope by at most 14%. Of 220 weekly association tests across indicators and cohorts, 205 remain significant after false-discovery-rate correction using the Benjamini–Hochberg procedure. Alternative bounded-outcome specifications agree in sign with the main models in all five cohorts, and varying the at-risk engagement threshold shifts recall and precision in the expected direction without changing the overall interpretation.

7 Discussion

Taken together, the results support the chapter-aligned engagement metric as a moderate but fairly stable observational proxy for academic performance in the STAT0002 and STAT0004 cohorts studied here. The pooled association is positive in every delivery, emerges early in most cohorts, and survives programme adjustment and false-discovery-rate correction using the Benjamini–Hochberg procedure. This external validation on new module-year contexts is broadly consistent with the primary findings of Johnston et al. (2025), while extending them through pooled inference and richer diagnostic analysis.

The most practically important pattern is that engagement appears especially informative for lower-performing students. Quintile separation, quantile regression and trajectory analysis all point in the same direction: students with sustained low or narrow engagement profiles are disproportionately represented among weaker grade outcomes. That does not mean the metric "predicts failure" with high accuracy—precision remains modest—but it does suggest value for early, low-cost support aimed at students whose engagement profile indicates they may be falling behind.

The diagnostic extensions add nuance that a single composite score alone would hide. Raw session frequency is the least informative of the three core indicators once collinearity is accounted for; breadth of resource use and promptness matter more. Temporal regularity adds further information in STAT0002, and trajectory shape separates outcomes as strongly as any single summary statistic. Engagement also aligns more closely with individually assessed work than with peer- or group-marked components, which strengthens the interpretation that Moodle logs capture individual study effort rather than collaborative performance.

Delivery condition does not provide a simple explanation for differences across cohorts. Descriptively, the engagement–grade relationship appears broadly similar across the online, hybrid and in-person periods represented in the data, but the design cannot disentangle delivery from year, cohort or curricular change. Module design, assessment structure and how clearly Moodle resources map to chapters are likely at least as important as the delivery label itself—a point reinforced by the stronger alignment with individual assessment components in STAT0002 and the weaker temporal-feature gains in STAT0004.

8 Practical implications

The metric is best understood as a transparent support tool rather than a high-stakes predictor. It can be monitored from early in the term, explained to students and instructors through its Frequency, Immediacy and Diversity components, and used to prompt low-cost interventions such as nudges, check-ins or invitations to office hours. Because many flagged students will ultimately pass, it should be used as one signal among several, not as a basis for punitive action.

When engagement is low, the more useful question is often *why* rather than *whether*: is the student behind the release schedule (Immediacy), touching only a narrow set of resources (Diversity), or following a persistently low trajectory? Those distinctions matter for how support is offered. Any operational use in a new module or delivery context should be preceded by local validation, especially where assessment design or resource labelling differs from the cohorts studied here.

9 Limitations

Several limitations should be kept in view. Delivery condition is confounded with academic year and cohort, so no causal inference about online, hybrid or in-person delivery is possible. The analysis covers only two statistics modules and five deliveries, limiting generalisability. Moodle logs capture online behavioural engagement only; offline study is invisible. Final grades are imperfect outcome measures, especially where group, peer or non-invigilated assessment contributes. Low-performer counts are small in some cohorts, making early-warning metrics unstable. The metric depends on consistent chapter labelling in Moodle. All findings are associational, not causal. Frequency, Immediacy and Diversity are highly collinear, so indicator-level interpretation requires appropriate methods. Finally, external validation would be needed before operational deployment in new contexts without recalibration.

10 Conclusion

This report provides secondary validation and diagnostic extension of an existing chapter-aligned Moodle engagement metric across STAT0002 and STAT0004 cohorts. Cumulative engagement is moderately and robustly associated with final grade, especially among lower-performing students. The most informative signals are breadth of resource use, promptness, temporal regularity and sustained trajectory pattern, rather than raw activity volume alone. Delivery-condition comparisons remain descriptive because condition is confounded with academic year and cohort. The metric is therefore best understood as a transparent tool for early monitoring and student support, not as a causal or high-stakes predictor of academic outcomes.

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